INCH-POUND

MIL-PRF-1/1100H
22 October 2019
SUPERSEDING
MIL-PRF-1/1100G
25 February 2005

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON
TYPES 5949A AND 5949A-1

Inactive For New Design
after 12 April 2012

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and MIL-PRF-1.

DESCRIPTION: Triode, hydrogen
See figures 1, and 2.
Mounting position: Any
Weight: 2.25 pounds (1.0 kg) nominal

RATINGS:

Parameter:	Ef	epy	epx	Ebb	Ip	egy	egx	Ecc	ib
Units:	V ac	kv	kv	Kv dc	A ac	v	v	V dc	a
Maximum:	6.6	25.0 1/	25.0 2/	---	15	---	450	---	500
Minimum:	6.0	10.0	5% epy	5.0	---	3/	---	---	---
Test conditions:	6.3	25.0	---	---	---	450	---	0	---
Parameter:	Ib	prr	tj	pb	tk	TA	dik/dt	Eres	Ires
Units:	A dc		μs			°C	a/μs	V ac	A ac
Maximum:	0.5	---	0.005 4/	6.25×10^9	---	+75	2,500	5.5 5/	6.0
Minimum:	---	---	3/	---	300	-55	---	3.0	---
Test conditions:	---	500	---	---	300	---	---	6/	---

1/ Instantaneous starting is not recommended. When it is absolutely necessary, however, the maximum permissible epy is 18.0 kv and shall not be attained in less than 0.04 seconds.
2/ The peak inverse voltage, exclusive of a spike of 0.05 μs maximum duration, shall not exceed 5 kv during the first 25 μs following the anode pulse.
3/ The driver pulse, measured at the tube socket with thyratron grid disconnected, shall be: egy = 550 volts minimum, 1,000 volts maximum; tr = 0.25 μs maximum; tp = 2.0 μs minimum; impedance of driver circuit 50 to 200 ohms.
4/ Appreciably less jitter than 0.005 μs can be realized if anode voltage of 18 kv or more, maximum rated grid drive voltage and a grid impedance near the lower limit are used.
5/ The optimum reservoir voltage for operation in accordance with operation (1) conditions is inscribed on the base of the tube and must be held to within ±5 percent.
6/ Adjust reservoir voltage to value indicated on tube to within ±5 percent.

GENERAL:
Qualification - Required

This specification sheet uses accept on zero defect sampling in accordance with MIL-PRF-1, table III.

Service-life guarantee – 500 hours minimum tube operating time.

Marking – Tubes sold under service-life guarantee shall be marked with the contract number and with the number of tube operating hours guaranteed.

Tube type 5949A superseded by 5949A-1.

AMSC N/A

FSC 5960

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:09Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection

Inspection	MIL-STD-1311 Method	Conditions	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 1						
Heater current	3241		If	15.0	22.0	Aac
Heater current	3241	Eres = 4.5 Vac	Ires	2.0	6.0	Aac
DC anode voltage for conduction	3247	Eres = inscribed value -5 percent	Ebb	---	4000	Vdc
Instantaneous starting 1/ 4/	3267	epy = 18.0 kv(min); Eres = inscribed value +5 percent	---	---	---	---
Operation (1) 1/	3246	epy = 27.5 kv; prr = 450;Eres = inscribed value -5 percent; t = 30 minutes	egy	---	450	v
Operation (2) 5/ 6/	3246	epy = 20 kv; prr = 1,200; Eres = 95 percent of optimum value (2)	egy	---	450	v
Operation of optimum reservoir voltage(1) (inscribed value)	3246	Operation (1)	Eres	3.0	5.5	Vac
Operation of 6/ optimum reservoir voltage (2)	3246	Operation (2)	Eres	3.0	5.5	Vac
Pulse emission	3251	ik = 500 a; tp = 5.0 μs ±10 percent; prr = 60 ±10 percent; tr = 0.5 μs (max); Eres = inscribed value; specified time interval = 2.5 μs	egk	---	250	v
Conformance inspection, part 2						
Operation (3)	3246	Operation (1), except Eres = inscribed value +5 percent; tk = 900 sec; t = 1,200 sec	egy	----	450	V
Operation (4)	3246	Operation (2), except Eres = 105 percent optimum value(2)	egy	----	450	V
Anode delay time	3256	Operation (1); t = 120 sec	tad	----	1.0	μs
Anode delay time drift 7/	3256	Anode delay time	Δtad	----	0.25	μs
Time jitter 8/	3261	Operation (1); except epy = 15.0 kv	tj	----	0.004	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:09Z
Check the source to verify that this is the current version before use.

TABLE I. Testing and inspection - Continued.

Inspection	MIL-STD-1311 Method	Conditions	Symbol	Limits Min	Limits Max	Unit
Conformance inspection, part 3						
Life test (1) 1/ 9/	----	Group c; t = 96 hours "on" and 1 hour "off"; t = 500 hours	----	----	----	----
Life test (2) 6/ 9/	----	Life test (1); operation (2); Zg = 250 ohms; t = 500 hours	----	----	----	----
Life test (3) 1/ 3/ 9/	----	t = 500 hours	----	----	----	----
Life-test (1), (2), and (3) end points:	----					
Operation (1)	3246		egy	----	500	V
DC anode voltage for conduction	3247	egy = 500 V	Ebb	----	4,500	V dc
Time jitter	3261	egy = 500 V; Zg = 250 ohms	tj	----	0.005	μs
Operation at elevated ambient temperature	3246	TA = 75°C; t = 5 hours	----	----	----	----

1/ The circuit constants shall be chosen so that at epy = 25.0 kv under charging conditions: dik/dt = 2,500 a/μs minimum; ib = 500 a minimum; tp = 2.0 ±0.2 μs; prr = 500 minimum. The grid pulse characteristics shall be: tp = 2.0 μs maximum; tr = 0.25 μs minimum; internal impedance of driver = 250 ohms minimum. With qualifying activity approval occasional shutdowns for testing or plant closing to cover off hours is permissible under life test.

2/ This test shall be conducted for a total of 5 consecutive hours with no more than three kickouts and with no evidence of anode heating. The tube shall be started at Eres = 105 percent of the inscribed value and operate at this value for 4 hours. At the start of the fifth hour and while the tube is still operating, the reservoir voltage shall be lowered to 95 percent of the inscribed value and remain there for the final 1 hour of operation.

3/ The life test shall be intermittent, with the tube operating approximately half the total elapsed time, and held without reservoir voltage or cathode-heater voltage approximately half the total time. The tube shall make a minimum of four starts during each 24-hour period. The life-test duration is measured in operating hours. Failure of the tube to operate following the minimum allowable cathode and reservoir heater warmup time constitutes failure of the test. With qualifying activity approval occasional shutdowns for testing or plant closing to cover off hours is permissible under life test.

4/ The tube shall operate satisfactorily on push-button starting within three attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 18 kv within 0.03 second. (The filter in the rectifier shall be designed so that the epy reaches at least 9 kv within 0.015 second.) Any tube failing to start within three attempts will be considered a failure.

5/ The anode circuit constants shall be so chosen that at resonant charging conditions; epy = 20 kv; ib = 200 a minimum; dik/dt = 2,500 a/μs minimum. Output pulse; tp = 1.0 μs ±10 percent; prr = 1,200 minimum; epx = 5.0 kv. Grid pulse the same as note 1. Adjust reservoir voltage to optimum value for these test conditions. The tube shall operate continuously for 20 minutes without continuous conduction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:09Z
Check the source to verify that this is the current version before use.

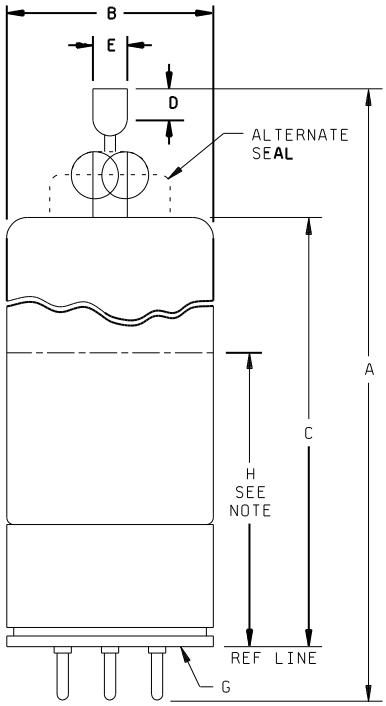
6/ The optimum reservoir voltage for operation in accordance with operation (2) conditions shall be so determined that the tube shall operate at 95 to 105 percent of the established reservoir voltage for these conditions.

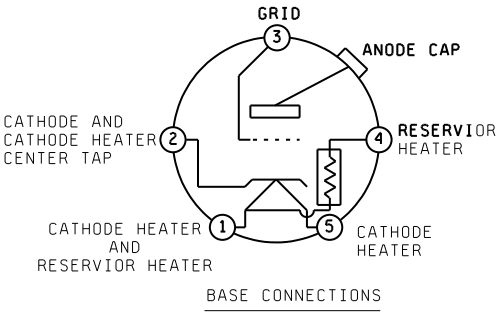
7/ This test shall be performed simultaneously with the operation (1) test. An anode delay time measurement shall be made at the end of 2 minutes, 4 minutes, and 30 minutes of the operation (1) test. The Δtad measurement is the numerical difference between the 2-minute and 4-minute, or 2-minute and 30-minute anode delay time readings, whichever is greater.

8/ The tj shall be not greater than the amount specified after 120 seconds of anode operation.

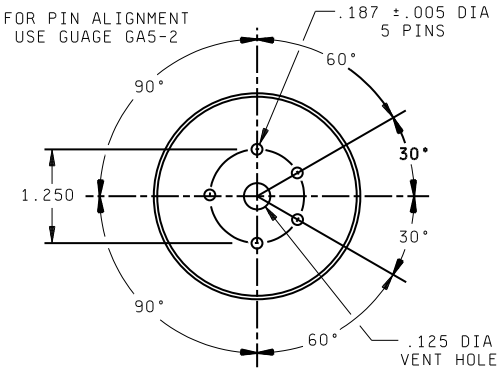
9/ With qualifying activity approval the manufacturer may provide, in accordance with MIL-PRF-1, service-life guarantee, in lieu of performing life testing. Life test endpoints specified shall apply to service-life guarantee conformance as well as life test conformance. The number of hours of system-deployed, accumulated tube-operating time shall be approved by the qualifying activity and shall be a minimum of 500 hours. Service-life guarantee shall define tube operating time and not time from purchase or delivery. Tubes sold under service-life guarantee shall be marked with contract number and with the number of tube operating hours guaranteed. The qualifying activity may restore life testing requirements when service performance or tube reliability indicate it is justified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:09Z
Check the source to verify that this is the current version before use.





BASE CONNECTIONS

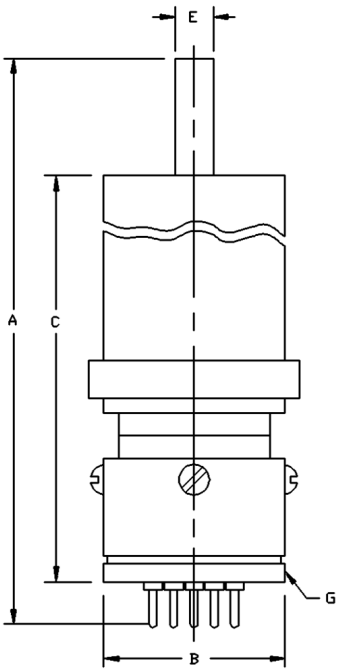


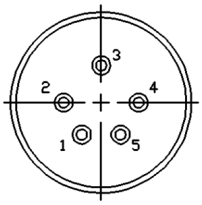
Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	11.500	12.500	292.10	317.50
B	--------	3.313	--------	84.15
C	8.000	9.000	203.20	228.60
H	--------	5.000	--------	127.00
Conformance inspection, part 3				
E	0.559	0.573	14.20	14.55
G	Base: A5-98 (EIA)			
Reference dimensions				
D	0.500		12.70	

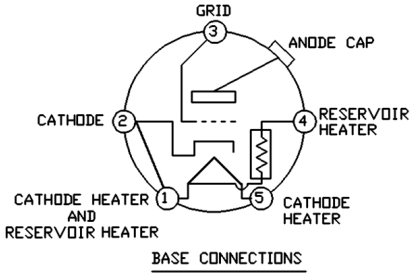
NOTES: Dimension H locates the center line of the clamping area. Clamping shall be confined to 1.5 inches above and below this line.

FIGURE 1. Outline drawing of electron tube type 5949A (glass)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:09Z
Check the source to verify that this is the current version before use.







Ltr	Dimensions			
	Inches		Millimeters	
Conformance inspection, part 2				
	Min	Max	Min	Max
A	11.500	12.500	292.10	317.50
B	-----	3.313	-----	84.15
C	-----	7.000	-----	177.80
E	0.559	0.573	14.20	14.55
G	Base: A5-98 (modified)			

NOTE: Five (5) pins are correct for 5949A and 5949A-1. The four (4) slots are for stress relief when epoxied to 5949A glass thyratron. “Modified” indicates the slots are replaced by through holes for screw fastening to metal-ceramic 5949A-1 thyratron. This difference is transparent to the end-user.

FIGURE 2. Outline drawing of electron tube type 5948A-1 (ceramic).

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:09Z
Check the source to verify that this is the current version before use.

Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-STD-1311

Revision notations. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2019-014)

Review activities:
Army - AR, AV, MI
Navy - AS, CG, MC, OS, SH
Air Force - 11

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:09Z
Check the source to verify that this is the current version before use.